ProteinNetworkSight Tutorial

Tutorial TOC

Summary

ProteinNetworkSight is a web-based tool for constructing patient-specific protein interaction networks from user-provided gene or protein scores. The platform integrates quantitative scores with curated interaction databases, applies network propagation to prioritize functionally relevant proteins, and visualizes the resulting networks as further linked to the Cancer Dependency Map, enabling the exploration of potential therapeutic targets.

ProteinNetworkSight doesn't track user data or activity, requires no login and is suitable for free use.

Cited Source Cellular Resolution:

ProteinNetworkSight integrates information from two external databases:

- STRING db - a database of known and predicted protein-protein interactions.
- Cancer Dependency Map - a resource combining tumor genetic dependencies.
- cellxgene - a protein sequence and functional information resource.

The platform combines these information sources to prioritize interactions and facilitate the identification of biologically relevant proteins and potentially drug-targetable proteins.

Workflow Overview

The ProteinNetworkSight analysis consists of three main steps:

1. Upload a gene or protein score file. The platform accepts common gene identifiers together with numeric scores (e.g., FC, p-values, fold changes, or other quantitative metrics).
2. Define analysis parameters - Users specify the identifier column, numeric column (prefix), interactor score threshold, and network propagation settings.
3. Explore the resulting networks - The platform constructs a protein interaction network using interactions data from the STRING database. Users can interactively explore the network, adjust visualization settings, and export the results in graphical or tabular formats.

Starting Point

To begin this analysis, users must upload an Excel, CSV or TSV containing a single sheet of tabular data. Each row should correspond to a protein or gene, and each column should represent a feature. The file must contain no more than 5000 rows. The column should contain protein or gene identifiers, while additional columns should contain numeric values representing gene/protein scores or quantitative metrics.

Note: Only the first numeric column (after column header) is currently considered for analysis (multi-column support is planned).

An example file is provided for download from Supplement 4 of Ng2 Nature Oncology 2023, Supplementary Data 10. In this example, the gene identifier column is labeled "GC", and the numeric column is labeled "FC with prefix "LC". Advanced by an error.

In this dataset, the 10 columns represent protein weights derived from the information theoretic analysis described in Sebastian et al. Nature Ecol. Evolution, the Fold or Rankle, and these columns has similar (any quantitative scores, coefficients, or Fold change ratios, depending on the method used to identify or expression patterns).

GC	FC	LC	Gene
1	0.0000000000	0.0000000000	0.0000000000
2	0.0000000000	0.0000000000	0.0000000000
3	0.0000000000	0.0000000000	0.0000000000
4	0.0000000000	0.0000000000	0.0000000000
5	0.0000000000	0.0000000000	0.0000000000
6	0.0000000000	0.0000000000	0.0000000000
7	0.0000000000	0.0000000000	0.0000000000
8	0.0000000000	0.0000000000	0.0000000000
9	0.0000000000	0.0000000000	0.0000000000
10	0.0000000000	0.0000000000	0.0000000000

Setting parameters and thresholds

In this step the user should specify:

1. Identifier column - The header name of the protein/gene identifier column (e.g., "GC" in the example file).
2. Numeric column prefix - The prefix shared by the numeric columns (e.g., "G", where G represents the weight of the protein in the corresponding cell line).
3. STRING interactor threshold - The score limit for the STRING interaction score, representing the probability of interaction between pairs of proteins.
4. Network propagation - Whether to propagate thresholds used to exclude proteins whose scores fall within a specified range.
5. Organism of interest - For example, Homo sapiens in the example dataset.

Manual thresholds adjustments

You can also choose to adjust the thresholds of every score (G in the example) column in the graph.

Adjusting proteinligene names

ProteinNetworkSight utilizes interactions data from STRING db, in order to construct and visualize protein interaction networks. Therefore, protein or gene identifiers in the input file should match the names used in STRING db. The platform supports common gene/protein identifiers, including gene symbols, UniProt IDs, and others automatically. The following steps highlight our cases and allow users to select alternative identifiers.

In step 2, The system suggests alternative identifiers recognized by the STRING database. For each gene, user can choose one of the suggested identifiers, or keep the original name as alternative identifier.

Gene/Protein Identifier Mapping
Click to Gene/protein identifier mapping
STRING Protein Identifier Mapping
Suggestions
Suggestions
Suggestions